Fixed-Misspecification RMSEA Bias: A Curvature Check

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Purpose

This note checks the fixed-misspecification version of the RMSEA bias calculation. The local-drift story gives the familiar chain

$$df \quad \longrightarrow \quad \mathrm{tr}(U_M \Gamma),$$

where U_M is the first-order residual projector and Γ is the asymptotic covariance of the saturated moments. Under fixed model misspecification this is not the full n^{-1} bias of the fitted discrepancy. The population residual is then $O(1)$, so two terms that are invisible under exact fit or Pitman drift move to leading bias order:

model curvature against the residual, data-dependent weight influence.

There is also a mundane but important bookkeeping term: if the first-stage moment estimator has an $O(n^{-1})$ mean bias, that bias contributes at the same order because the population residual is nonzero.

The result below is therefore the fixed-misspecification correction

$$\mathbb{E}[\hat{F}] = F_0 + \frac{1}{n}\left\{\mathrm{tr}(Q_M \Gamma) + \beta_W + 2e'Wb\right\} + o(n^{-1}),$$

with Q_M built from the observed population Hessian, not just the Gauss–Newton Hessian. For a fixed weight and an unbiased first-stage covariance convention this reduces to $\mathrm{tr}(Q_M \Gamma)$.

The notation is deliberate. U_M is reserved for the exact-fit or Gauss–Newton residual projector. Q_M denotes the profile Hessian of the fitted value function. For an absolute-fit statistic, the comparator is the saturated model, whose profile value is identically zero, so the quadratic form is $Q_M - Q_{\mathrm{sat}} = Q_M$. For a nested test, the corresponding object is $Q_B - Q_A$.

1 Setup

Let m be the saturated moment vector used by the estimator. For ordinary complete-data covariance SEM, $m = \mathrm{vech}(\Sigma)$; for other settings it may include means, thresholds, polychorics, or group blocks. Assume the sample version has the expansion

$$\hat{m} = m_0 + n^{-1/2} z + n^{-1} b_n + o_p(n^{-1}), \qquad z \xrightarrow{d} N(0, \Gamma), \qquad \mathbb{E}[b_n] \to b. \quad (1)$$

The vector b is zero for an exactly unbiased first-stage convention, but it is not automatically zero for every covariance divisor or saturated statistic.

Let the model be a smooth map $\theta \mapsto \mu(\theta)$ and define the quadratic discrepancy

$$\hat{F} = \min_{\theta} r_n(\theta)' \hat{W} r_n(\theta), \qquad r_n(\theta) = \hat{m} - \mu(\theta). \quad (2)$$

The pseudo-true point for the limiting weight W is

$$\theta_* = \arg\min_{\theta} \{m_0 - \mu(\theta)\}' W \{m_0 - \mu(\theta)\}, \qquad e = m_0 - \mu(\theta_*), \qquad F_0 = e' W e. \quad (3)$$

Write $D = \partial \mu / \partial \theta'$ at θ_*. The population first order condition is

$$D' W e = 0. \quad (4)$$

Let H_a be the Hessian matrix of the ath component of $\mu(\theta)$ at θ_*. Define

$$A = D' W D, \qquad K_{ij} = \sum_a (We)_a (H_a)_{ij}, \qquad B = A - K. \quad (5)$$

Here A is the Gauss–Newton part and B is one half of the population Hessian of $e'We$ with respect to θ. The residual-curvature term K vanishes under correct specification and for a locally linear model. Under fixed misspecification it is generally not negligible.

For an estimated weight, write

$$\hat{W} = W + n^{-1/2} W_1 + n^{-1} W_2 + o_p(n^{-1}), \quad (6)$$

with W_1 the mean-zero first-order influence of the weight map. Put

$$g = W_1 e, \qquad P = D B^{-1} D', \qquad Q_M = W - W D B^{-1} D' W = W - W P W. \quad (7)$$

2 Expansion

Theorem 1 (Fixed-misspecification expansion)**.** *Assume* (1)–(6)*, smoothness, full column rank D, and nonsingular B. Then*

$$\hat{F} = F_0 + n^{-1/2}\{2e'Wz + e'W_1e\} + n^{-1}\Phi + o_p(n^{-1}), \tag{8}$$

where

$$\Phi = z'Q_Mz + 2g'(I - PW)z - g'Pg + e'W_2e + 2e'Wb_n. \tag{9}$$

Proof. Let $\hat{\theta} - \theta_* = n^{-1/2}\xi + O_p(n^{-1})$. Expanding the estimating equation $D(\hat{\theta})'\hat{W}\{\hat{m} - \mu(\hat{\theta})\} = 0$ to first order and using $D'We = 0$ gives

$$(A - K)\xi = D'(Wz + W_1e), \qquad \xi = B^{-1}D'(Wz + g).$$

Now expand the fitted residual:

$$r_n(\hat{\theta}) = e + n^{-1/2}(z - D\xi) + n^{-1}\{b_n - \eta - \tfrac{1}{2}H[\xi,\xi]\} + o_p(n^{-1}),$$

where η is the second-order parameter increment and $H[\xi,\xi]$ stacks $\xi'H_a\xi$. In $r_n(\hat{\theta})'\hat{W}r_n(\hat{\theta})$, the population FOC kills the terms involving η and leaves

$$z'Wz - 2\xi'D'Wz + \xi'B\xi + 2g'(z - D\xi) + e'W_2e + 2e'Wb_n.$$

Substituting $\xi = B^{-1}D'(Wz + g)$ and collecting terms gives (9). □

Corollary 1 (Bias)**.** *If* $\mathbb{E}[W_1] = 0$ *and the expectations below exist, then*

$$\mathbb{E}[\hat{F}] = F_0 + \frac{1}{n}\left[\mathrm{tr}(Q_M\Gamma) + \beta_W + 2e'Wb\right] + o(n^{-1}), \tag{10}$$

where

$$\beta_W = 2\mathbb{E}\{g'(I - PW)z\} - \mathbb{E}(g'Pg) + e'\mathbb{E}(W_2)e. \tag{11}$$

For a fixed weight, $W_1 = W_2 = 0$ and $\beta_W = 0$. If additionally $b = 0$, the whole correction is just

$$\mathbb{E}[\hat{F}] = F_0 + n^{-1}\,\mathrm{tr}(Q_M\Gamma) + o(n^{-1}).$$

Under correct specification, $e = 0$, so $K = 0$, Q_M reduces to the usual first-order residual projector

$$U_M = W - WD(D'WD)^{-1}D'W,$$

and the bias reduces to $n^{-1}\,\mathrm{tr}(U_M\Gamma)$. Under normal-theory weighting with $W = \Gamma^{-1}$, this trace is df.

3 Consequences for RMSEA

The fixed-population RMSEA target is

$$\varepsilon_0 = \sqrt{F_0/df}.$$

When $F_0 > 0$ is fixed, the leading distribution of $\hat{F}$ is not noncentral chi-square. It is ordinary root-n normal, because the linear term in (8) is nonzero:

$$\sqrt{n}(\hat{F} - F_0) \xrightarrow{d} N(0, \Omega_F), \qquad \Omega_F = \mathrm{Var}(2e'Wz + e'W_1e). \tag{12}$$

The n^{-1} bias correction improves centering but does not change the root-n variance. By the delta method,

$$\sqrt{n}(\hat{\varepsilon} - \varepsilon_0) \xrightarrow{d} N\left(0, \frac{\Omega_F}{4df^2\varepsilon_0^2}\right), \qquad F_0 > 0.$$

Thus the noncentral-χ^2 interval belongs to the exact-fit/local-drift regime. Under fixed misspecification, a Wald interval based on (12) is the natural large-sample object.

On the point-estimation scale, write

$$c_{\text{fix}} = \mathrm{tr}(Q_M\Gamma) + \beta_W + 2e'Wb.$$

Then a fixed-misspecification bias-corrected RMSEA estimator is

$$\hat{\varepsilon}_{\text{fix}} = \sqrt{\max\left(\frac{\hat{F} - c_{\text{fix}}/n}{df}, 0\right)}. \tag{13}$$

Equivalently, if $T = n\hat{F}$, subtract c_{fix} from T, not automatically df or $\mathrm{tr}(U_M\Gamma)$.

Remark. The truncation in (13) is asymptotically irrelevant when $F_0 > 0$. It only protects finite samples and connects the formula back to the usual RMSEA convention near exact fit, where the limiting law changes from the normal law in (12) to a quadratic form.

4 Estimator Cases

ULS or fixed-weight GLS. This is the clean case. The weight is fixed, so $\beta_W = 0$. If the saturated moments are centered so that $b = 0$, the correction is $\mathrm{tr}(Q_M\Gamma)$. This is the most direct target for simulation checks: compare $n\{\mathbb{E}[\hat{F}] - F_0\}$ with both $\mathrm{tr}(U_M\Gamma)$ and $\mathrm{tr}(Q_M\Gamma)$.

Data-weighted GLS. The formula applies directly if $\hat{W}$ is a smooth function of $\hat{m}$ and does not also depend on θ. The new object is β_W, which is at least linear in the residual through $g = W_1 e$, with quadratic pieces through $g'Pg$ and $e'W_2e$. It vanishes at exact fit but is part of the same n^{-1} bias under fixed misspecification.

ML. Covariance ML is not simply a data-weighted quadratic form with a fixed θ-free weight. It can be locally represented by a quadratic form, but the weight is tied to the model-implied covariance at the fitted parameter. That means the same shape should reappear after a direct profile-likelihood expansion, but the GLS β_W term should not be copied over without doing that calculation. The safe next derivation is therefore ML-specific, not a blanket substitution into (11).

5 What This Says

The curvature correction is mathematically real. The fixed-misspecification bias uses the profile Hessian $B = A - K$, so the trace is $\mathrm{tr}(Q_M\Gamma)$ rather than $\mathrm{tr}(U_M\Gamma)$ unless either the residual or the model curvature is negligible. This is the same phenomenon as replacing expected information by observed Hessian in misspecification-robust standard errors.

The $O(n^{-1})$ moment-bias term should be explicit. It is absent only after a choice of first-stage convention, not as a theorem. For example, an unbiased sample covariance convention removes it, while a different divisor may not.

The most useful next step is a small, fast validation harness for the fixed ULS case. Once $\mathrm{tr}(Q_M\Gamma)$ is verified there, add controlled data-weighted GLS examples to isolate β_W, and derive ML separately from the profiled likelihood. That sequencing keeps the algebra testable: first curvature, then weight influence, then likelihood-specific cancellation.